Ouail Zakary
Postdoctoral Researcher, Ph.D. in Physics
NMR Research Unit, Faculty of Science, University of Oulu
P.O. Box 8000
FIN-90014 University of Oulu
Finland
E-mail: ouail.zakary@oulu.fi
Personal Website: https://ozakary.github.io
Website: Ouail Zakary
GitHub: ozakary
Scopus ID: 58662986700
ResearcherID: OSI-7733-2025
ORCID: 0000-0002-7793-3306
Google Scholar: Ouail Zakary

List of Publications

List date: September 7, 2026
Note: Corresponding authorship is marked with (*)

A. Peer-reviewed scientific articles

A.1. Published articles

1. **Zakary, O.***; Body, M.*; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Structural Modeling of O/F Correlated Disorder in $TaOF_3$ and $NbOF_{3\text{-}x}(OH)_x$ by Coupling Solid-State NMR and DFT Calculations. *Inorg. Chem.* **2023**, *62*, 16627–16640.
[DOI: 10.1021/acs.inorgchem.3c02844] [Supporting Data]

2. **Zakary, O.***; Body, M.; Sarou-Kanian, V.; Arnaud, B.; Corbel, G.*; Legein, C. Different Magnitudes of Second-Order Jahn-Teller Effect in Isostructural $NaMO_2F_2$ (M= Nb, Ta) Oxyfluorides. *J. Alloys Compd.* **2025**, *1010*, 177457.
[DOI: 10.1016/j.jallcom.2024.177457] [Supporting Data] [Supporting Code]

3. **Zakary, O.***; Body, M.; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Revealed Preferential Short-Range Anion Ordering in Disordered RbM_2O_5F (M= Nb, Ta) Pyrochlore-Type Oxyfluorides. *Inorg. Chem.* **2025**, *64*, 5764–5777.
[DOI: 10.1021/acs.inorgchem.5c00615] [Supporting Data] [Supporting Code]

4. **Zakary, O.***; Lantto, P. Equivariant Neural Networks Reveal How Host–Guest Interactions Shape ^{129}Xe NMR in Porous Liquids. *J. Phys. Chem. Lett.* **2025**, *16*, 12095–12103. (Selected as *Journal Supplementary Cover*)
[DOI: 10.1021/acs.jpclett.5c02846] [Supporting Data] [Supporting Code] [Journal Cover]

5. Laurila, O.; Jacklin, T.; **Zakary, O.***; Lantto, P.* Machine Learning-Accelerated Path Integral Molecular Dynamics and ^{13}C NMR Simulations Unlock New Insights Into Quantum Effects in C_{60} Fullerene. *J. Phys. Chem. A* **2026**, 130, 2169–2181.
[DOI: 10.1021/acs.jpca.6c00238] [Supporting Data] [Supporting Code]

A.2. Pre-prints (submitted articles/under review)

6. **Zakary, O.***; Jacklin, T.; Lantto, P. Carbon Nanotube-Induced Magnetic Shielding Effects on ^{129}Xe NMR from Equivariant Neural Networks. *ChemRxiv* **2026**, *under review in Phys. Chem. Chem. Phys.*
[DOI: 10.26434/chemrxiv.15007956/v1] [Supporting Data] [Supporting Code]

7. **Zakary, O.***; Yin, W.; Aryal, N.* Local Symmetry Breaking and Correlated Fluctuations in Quantum Materials from Atomistic Foundation Models. *ChemRxiv* **2026**, *under review in npj Comput. Mater.*
[DOI: 10.26434/chemrxiv.15001387/v2] [Supporting Data] [Supporting Code]

A.3. Articles in preparation

8. **Zakary, O.***; Jacklin, T.; Lantto, P. Machine Learning-Powered Xenon NMR Relaxation in Carbon Nanotubes. 2026, *In preparation.*

9. **Zakary, O.***; Body, M.; Sarou-Kanian, V.; Legein, C. Hidden Short-Range Order in Mixed-Anion Sodium-Oxyfluorotantalates. 2026, *In preparation.*

10. Laurila, O.*; **Zakary, O.***; Lantto, P.* Machine Learning for Computing NMR Observables Highly Sensitive to Quantum Effects. 2026, *In preparation.*

11. Laurila, O.*; **Zakary, O.***; Lantto, P.* ^{3}He NMR Monomer-Dimer Shift in $He_m@C_{60}^{n-}$ and $He_m@C_{70}^{n-}$ ($m = 1, 2$, and $n = 0, 6$) Endohedral Fullerenes from Machine Learning-Guided Simulations. 2026, *In preparation.*

12. **Zakary, O.***; Mailhiot, S.E.; Selent, A.; Mareš, J.; Lantto, P.*; Telkki, V.-V.* Unifying the Description of Type-II Porous Liquids Through Machine Learning-Accelerated Molecular Dynamics and ^{129}Xe NMR: Simulations Informing Experiments. 2026, *In preparation.*

13. Hintsanen, M.; Lantto, P.; **Zakary, O.*** Insights into the Density Maximum of Water from Machine Learned ^{129}Xe NMR and Molecular Dynamics. 2027, *In preparation.*

14. **Zakary, O.*** A Machine Learning Workflow for Solving Disorder in Heteroanionic Materials. 2027, *In preparation.*

15. **Zakary, O.*** A Foundation Model for NMR Parameters in Materials Chemistry. 2027, *In preparation.*

16. Laurila, O.*; **Zakary, O.***; Lantto, P.* Machine Learning-Driven Path Integral Molecular Dynamics and ^{13}C NMR Simulations Enable the Assignements of Singlet NMR in the Case of High Molecular Symmetry. 2027, *In preparation.*

17. Selent, A.; **Zakary, O.***; Mareš, J.; Lantto, P.*; Telkki, V.-V.* Probing R3@DMSO and R3S@H2O Porous Liquids Combining ^{129}Xe NMR and Machine Learning-Accelerated Simulations. 2027, *In preparation.*

18. Chennabassapa, M.; Squires, A.G.; Demourgues, A.; Penin, N.; Durand, E.; Li, W.; Legein, C.; **Zakary, O.**; Body, M.; Charpentier, T.; Borkiewicz, O.J.; Morgan, B.J; Scanlon, D.O.; Dambournet, D.* Hexagonal-Tungsten-Bronze $TiOF_2$: Synthesis, Crystal Structure and Anionic Short-Range Ordering. 2027, *In preparation.*

B. Non-refereed scientific articles

B.1. Conference proceedings and presentations

B.1.1. Oral presentations

1. **Zakary, O.**; Body, M.; Charpentier, T.; Legein, C. Structural Modeling of Disordered Inorganic Oxyfluorides by Coupling Solid State NMR and DFT Calculations. *Réunion RMN Grand Bassin Parisien*, 19 Nov. **2021**, Orsay, France.
[DOI: 10.6084/m9.figshare.27111955.v2]

2. Legein, C.; Body, M.; **Zakary, O.**; Lhoste, J; Fayon, F.; Dambournet, D. Caractérisations et modélisations structurales de fluorures inorganiques désordonnés : apports de la RMN du solide et des calculs DFT. *Colloque Français de Chimie du Fluor (CFCF)*, 16–19 May **2022**, Forges-les-Eaux, France.
[DOI: hal.science/hal-05287392v1]

3. **Zakary, O.**; Body, M.; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Structural Modeling of O/F Correlated Disorder in $NbOF_3$ and $TaOF_3$. *Alpine Conference on Magnetic Resonance on Solids*, 04-08 Sept. **2022**, Chamonix-Mont-Blanc, France.
[DOI: 10.6084/m9.figshare.27111496.v2]

4. **Zakary, O.** Towards a Complete Structural Description of Disordered Inorganic Fluorides: The Crucial Role of Solid State NMR and DFT Calculations. *JED – 3MG, Journée de l'Ecole Doctorale*, 25 May **2023**, Nantes, France.
[DOI: 10.6084/m9.figshare.27115360.v2]

5. **Zakary, O.**; Lantto, P. Modeling Porous Liquids with Machine-Learning-Assisted Molecular Dynamics. *Computational Chemistry Days (CCD)*, 27-28 May **2024**, Jyväskylä, Finland.
[DOI: 10.6084/m9.figshare.27105913.v1]

6. **Zakary, O.**; Lantto, P. Scalable, Accurate, and Data-Efficient Machine Learning Potential models for Realistic Simulations of Porous Materials. *Chemical Solutions for Biological Challenges (CSBC)*, 10-12 June **2024**, Turku, Finland.
[DOI: 10.6084/m9.figshare.27105937.v2]

7. **Zakary, O.**; Lantto, P. Efficient and Accurate Local Equivariant Deep Neural Network Interatomic Potential for Large-Scale Porous Liquids Simulations. *Magnetic Resonance in Porous Media 16 (MRPM)*, 26-30 Aug. **2024**, Tromsø, Norway.
[DOI: 10.6084/m9.figshare.27106033.v2]

8. **Zakary, O.**; Lantto, P. Efficient and Accurate Local Equivariant Deep Neural Network Interatomic Potential for Large-Scale Porous Liquids Simulations. *Winter School in Theoretical Chemistry*, 09-12 Dec. **2024**, Helsinki, Finland.
[DOI: 10.6084/m9.figshare.29757140.v1]

9. **Zakary, O.**; Lantto, P. Equivariant Neural Networks Reveal How Host–Guest Interactions Shape Xenon NMR Chemical Shift in Porous Organic Cages. *The 21th European Magnetic Resonance Congress (EUROMAR)*, 06-10 July **2025**, Oulu, Finland.
[DOI: 10.6084/m9.figshare.29757203.v1]

10. **Zakary, O.** Understanding Host-Guest Interactions in Type-II Porous Liquids using Machine Learning-Accelerated Molecular and NMR Parameters Simulations. *Computational Chemistry*

Days (CCD), 25-26 May **2026**, Helsinki, Finland.
[DOI: 10.6084/m9.figshare.32519391]

11. **Zakary, O.** Machine Learning-Accelerated Molecular Dynamics and NMR Parameter Simulations in Porous Materials. *SMARTER 9*, 30 Aug.-03 Sept. **2026**, Bayreuth, Germany.
[DOI: 10.6084/m9.figshare.33446857]

B.1.3. Poster presentations

12. **Zakary, O.**; Body, M.; Legein, C. Structural Modeling of Disordered Inorganic Fluorides. *JED – 3M445, Journée de l'Ecole Doctorale*, 27-28 May **2021**, Le Mans, France.
[DOI: 10.6084/m9.figshare.27115426.v1]

13. **Zakary, O.**; Lantto, P. Combining Molecular Dynamics with Deep Neural Network Architectures for Realistic Simulations of Porous Liquids. *The 20th European Magnetic Resonance Congress (EUROMAR)*, 30 June - 04 July **2024**, Bilbao, Spain.
[DOI: 10.6084/m9.figshare.27105562.v2]

14. **Zakary, O.**; Lantto, P. Machine Learning Driven Approach for Modeling Host-Guest Dynamics in Xenon-Based Porous Liquids. *Physics Days*, 26-28 Mar. **2025**, Oulu, Finland.
[DOI: 10.6084/m9.figshare.28684814.v1]

15. **Zakary, O.**; Lantto, P. Local E(3)-Equivariant Neural Network Force Field for Modeling Host-Guest Interactions in Xenon-Based Porous Organic Cages. *Computational Chemistry Days*, 02-03 June **2025**, Espoo, Finland.
[DOI: 10.6084/m9.figshare.28684814.v1]

16. **Zakary, O.**; Lantto, P. Host-Guest Dynamics in Porous Liquids Modeled Using E(3)-Equivariant Neural Networks. *The 13th Triennial Congress of the World Association of Theoretical and Computational Chemists (WATOC 2025)*, 21-27 June **2025**, Oslo, Norway.
[DOI: 10.6084/m9.figshare.28684814.v1]

17. **Zakary, O.**; Lantto, P. Machine Learning-Accelerated Molecular Dynamics and NMR Parameter Simulations in Porous Materials. *The 22st European Magnetic Resonance Congress (EUROMAR)*, 28 June - 02 July **2026**, Gothenburg, Sweden.
[DOI: 10.6084/m9.figshare.32895569]

B.1.4. Schools

18. Moving ions with VASP. *University of Vienna*, 27-30 Sept. 2022, Online (Zoom)

19. Machine Learning of First Principles Observables. *Centre Européen de Calcul Atomique et Moléculaire (CECAM)*, 08-12 July 2024, Berlin, Germany.

20. Electronic Structure Theory. *Winter School in Theoretical Chemistry*, 15-18 Dec. 2025, Helsinki, Finland.

C. Theses

C.1. Doctoral dissertation

- **Zakary, O.** Structural Modeling of Oxygen-Fluorine Ordering in Transition Metal Inorganic Oxyfluorides. *Le Mans Université* **2023**.
[Thesis] [Ph.D. Defense Details]